11-1. Matching

A consolidated, mathematical lecture companion

English notes based on the lecture slides *Matching* (Prof. Sanghack Lee, GSDS SNU; drawing on slides by Fan Li, Kosuke Imai and Jingshu Wang)

The central question is how to estimate the effect of treatment on the treated by imputing each treated unit's missing control outcome from similar control units, and how to design such comparisons so that the result does not hinge on modelling choices.

These notes consolidate the slides, work the nearest-neighbour toy example, explain the rate and inference results of Abadie and Imbens, and collect the practical workflow (distances, calipers, full matching, balance checks) and the empirical lessons. Physical PDF page numbers are listed in Section H.

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1 Matching as nonparametric imputation

Purpose. Motivate matching as a model-free way to impute counterfactual control outcomes.

1.1 Setup

N units, treatment $X_i \in \{0,1\}$, covariates Z_i, potential outcomes $Y_i(0), Y_i(1)$; $N_1 = |\{i : X_i = 1\}|$ treated, $N_0 = N - N_1$ controls. Assumptions: ignorability $(Y(0), Y(1)) \perp\!\!\!\perp X \mid Z$ and positivity $0 < P(X = 1 \mid Z) < 1$. Focus: the ATT.

1.2 From regression to matching

Under ignorability, with $\mu_t(Z) = \mathbb{E}[Y \mid X = t, Z]$,

$$\text{ATT} = \mathbb{E}_{Z|X=1}\big[\underbrace{\mu_1(Z)}_{\text{observed}} - \underbrace{\mu_0(Z)}_{\text{impute}}\big].$$

Regression performs model-based imputation of $\mu_0(Z_i)$:

$$\hat{\tau}_{\text{reg}} = \frac{1}{N_1}\sum_{i=1}^{N} X_i(Y_i - \hat{\mu}_0(Z_i)).$$

When the groups differ in Z, the estimate depends heavily on the functional form of $\hat{\mu}_0$ (model dependence, Section 5). Matching performs *nonparametric* imputation instead.

1.3 The matching idea

Often there is a large reservoir of potential controls, so each treated unit can be matched to one or more distinct controls with similar covariates.

Matching as study design. Matching is a design tool, not only an analysis technique. In many observational studies outcomes are not yet measured when matching is done: identify treated units, select which controls to recruit/follow, then measure outcomes for both. This separates the design stage (whom to compare) from the analysis stage (how to estimate); design choices are made without seeing outcomes, removing the temptation to tune the analysis toward a desired result (Stuart 2010, §1).

Bias–variance trade-off in M. Each treated unit may be matched to M controls. $M = 1$: lower bias, higher variance. $M = 3$: farther matches increase bias, while averaging three outcomes reduces variance (Abadie and Imbens 2006).

1.4 The matching estimator

Let $\mathcal{M}_i$ be the matched set of treated unit i (units with covariates similar to i's). Still assuming ignorability,

$$\hat{\tau}_{\text{match}} = \frac{1}{N_1}\sum_{i=1}^{N} X_i\Big(Y_i - \underbrace{\frac{1}{|\mathcal{M}_i|}\sum_{j\in\mathcal{M}_i} Y_j}_{\approx Y_i(0)}\Big).$$

Matching can be one-to-one or one-to-many, with or without replacement, etc.

1.5 Exact matching and coarsened exact matching

Exact matching: for each treated i find a control j with $Z_j = Z_i$. It ensures perfect balance and no model dependence but is usually infeasible with many or continuous covariates.
Coarsened exact matching (CEM; Iacus et al. 2012): discretise covariates (categories may have substantive meaning), then match exactly on the coarsened values. Some treated units may still have no matches (lack of overlap).

2 Nearest-neighbour matching and its theory

Purpose. Define NN matching, compare matching algorithms, and state its large-sample behaviour.

2.1 Definition

Let $\mathcal{M}_i$ be the indices of the M closest units of the opposite group in a distance $\|\cdot\|$. The imputed potential outcomes are

$$\hat{Y}_i(t) = \begin{cases} Y_i, & X_i = t, \\ \dfrac{1}{M}\sum_{j\in\mathcal{M}_i} Y_j, & X_i = 1-t, \end{cases} \qquad t \in \{0,1\},$$

and the ATT estimator is $\hat{\tau}_{\text{ATT}} = \frac{1}{N_1}\sum_{i=1}^{N} X_i\big(Y_i - \hat{Y}_i(0)\big)$.

2.2 Algorithms (matching without replacement)

- **Greedy:** order the treated units (e.g. hardest-to-match first); give each the M closest remaining controls, set them aside, repeat.
- **Optimal:** build the $N_1 \times N_0$ distance matrix D; choose entries so that each row has exactly M entries and each column at most one, minimising the total distance. The Hungarian algorithm solves this in polynomial time.

Why order matters. With $M = 1$ and the same data, an arbitrary order lets i_2 take the only control near the isolated i_3 (distance 0.9), forcing i_3 to a control 2.9 away (worst distance 2.9). Processing the hardest unit first, i_3 secures its nearby control (1.2), i_2 takes another at 1.1, and the worst distance drops to 1.2.

2.3 The curse of dimensionality

In one dimension N uniform points have spacing about $1/N$; in p dimensions about $N^{-1/p}$ per axis, so the nearest-neighbour distance is $O(N^{-1/p})$. If μ_0 is Lipschitz with constant L,

$$|\mu_0(Z_i) - \mu_0(Z_j)| \le L\,\|Z_i - Z_j\| = O(N^{-1/p}),$$

which is the bias from mismatch. The rate slows dramatically with p:

p	1	5	10	20
$N^{-1/p}$ for $N = 10{,}000$	0.0001	0.16	0.40	0.63

This motivates dimension reduction, e.g. propensity-score matching (Abadie and Imbens 2006, Thm. 1).

2.4 ATT matching: independence and consistency

For NN matching with fixed M and replacement, the ATT imputes $\hat{Y}_i(0)$ for each treated unit from controls; each treated unit appears exactly once as a subject, so the imputed counterfactuals are independent across treated units. Independence plus the law of large numbers (and the bias being negligible when the control reservoir is large) give $\sqrt{N}$-consistency for the ATT. Matching is most natural for ATT estimation (Abadie and Imbens 2006, Thms. 1–4).

2.5 A toy example

$N_1 = 3$ treated, $N_0 = 6$ controls, $p = 2$, $M = 1$, distance $\|\cdot\|_1$:

	Treated				Control		
Unit	Z_1	Z_2	Y	Unit	Z_1	Z_2	Y
$i=1$	3	5	8	$j=1$	1	2	2
$i=2$	6	2	12	$j=2$	2	6	4
$i=3$	7	7	10	$j=3$	4	1	6
				$j=4$	3	4	5
				$j=5$	5	3	7
				$j=6$	8	6	9

Matches: $i = 1 \to j = 4$ (distance 1), $i = 2 \to j = 5$ (2), $i = 3 \to j = 6$ (2). Hence

$$\hat{\tau}_{\text{ATT}} = \tfrac{1}{3}\big[(8-5) + (12-7) + (10-9)\big] = 3.0.$$

The matches are imperfect ($Z_2 - Z_5 \neq 0$), and the mismatch creates bias, which the next estimator corrects.

2.6 Bias-corrected matching (Abadie and Imbens 2011)

NN matching uses Y_j to estimate $Y_i(0)$, but Y_j approximates $\mu_0(Z_j)$, not $\mu_0(Z_i)$. Adjust each control outcome by the predicted difference due to mismatch:

$$\hat{\tau}_{\text{bc}} = \frac{1}{N_1} \sum_{i:X_i=1} \Big(Y_i - \frac{1}{M} \sum_{j \in \mathcal{M}_i} \big[Y_j + \hat{\mu}_0(Z_i) - \hat{\mu}_0(Z_j) \big] \Big),$$

where $\hat{\mu}_0$ is a regression fit of $\mathbb{E}[Y \mid Z, X = 0]$. Naive matching uses Y_j as $Y_i(0)$, ignoring the slope of μ_0 between Z_j and Z_i; the correction shifts by $\hat{\mu}_0(Z_i) - \hat{\mu}_0(Z_j)$. This is the default in the R packages `MatchIt` and `Matching`.

2.7 Inference after matching

Variance estimator for the ATT (Abadie and Imbens 2006):

$$\widehat{\text{Var}}(\hat{\tau}_{\text{ATT}}) = \frac{1}{N_1^2} \sum_{i=1}^{N} \big(1 + K_M(i)\big)^2 \hat{\sigma}_i^2,$$

where $K_M(i)$ is the number of times unit i is used as a match and $\hat{\sigma}_i^2$ estimates $\text{Var}(Y_i \mid Z_i, X_i)$. Units matched often inflate the variance, analogous to extreme IPW weights. **The bootstrap is invalid** for NN matching: bootstrap validity requires small data changes to cause small estimate changes, but the identity of the matched unit is a discontinuous function of the data (Abadie and Imbens 2008).

2.8 Kernel matching

All controls within a bandwidth h get positive weight, closer ones more. With a kernel K (e.g. Gaussian $K(u) = e^{-u^2/2}$),

$$w_i(j) = \frac{K\left(\frac{Z_j - Z_i}{h}\right)}{\sum_{k:X_k=0} K\left(\frac{Z_k - Z_i}{h}\right)}, \qquad \hat{\tau}_{\text{kernel}} = \frac{1}{N_1} \sum_{i=1}^{N} X_i \Big(Y_i - \sum_{j:X_j=0} w_i(j) Y_j\Big).$$

For large p kernel matching faces the same curse of dimensionality.

3 Propensity-score matching

Purpose. Reduce the matching problem to one dimension and understand the caveats.

Dimension reduction. NN bias grows like $O(N^{-1/p})$; with 20 binary covariates there are $2^{20} \approx 10^6$ patterns and direct matching is nearly impossible. The propensity score $e(Z) = P(X = 1 \mid Z)$ reduces p dimensions to one, and $Y(0), Y(1) \perp\!\!\!\perp X \mid Z$ implies $Y(0), Y(1) \perp\!\!\!\perp X \mid e(Z)$.

Distance. $d(Z_i, Z_j) = |\hat{e}(Z_i) - \hat{e}(Z_j)|$, or (preferred) on the logit scale

$$d(Z_i, Z_j) = |\operatorname{logit} \hat{e}(Z_i) - \operatorname{logit} \hat{e}(Z_j)|, \qquad \operatorname{logit}(p) = \ln \frac{p}{1-p}.$$

Distances that are small on the probability scale are compressed near 0 and 1; the logit stretches them out.

Specification: balance, not fit. The PS is usually estimated by logistic regression, but GBMs, random forests, etc. are also used. The goal is covariate balance, not accurate prediction of treatment. Collinearity is not a concern (coefficients are not interpreted); overfitting to the sample can even help (Appendix). Specify iteratively: check balance on covariates, squares and interactions; if imbalanced, add these terms and re-estimate. With missing covariate values, impute and include missingness indicators (Stuart 2010; Rosenbaum and Rubin 1984).

Common support. Each treated unit needs controls with similar covariates. Units outside the overlap region of the two PS distributions have no good matches and should be trimmed.

Caveats. Matching on the *estimated* PS can be more efficient than on the true PS (Abadie and Imbens 2016): estimation noise adapts to the sample, so matched units are closer in covariate space; this relies on correct specification. King and Nielsen (2019) describe the "PS paradox": a narrower PS caliper improves PS balance but can worsen covariate balance, because units with similar $\hat{e}(Z)$ can have very different Z. Always check multivariate covariate balance, not just PS balance.

4 Practical matching

Purpose. Describe the iterative matching workflow and its design choices.

Workflow. Choose a distance → perform matching → check balance → if imbalanced, adjust (metric, M, caliper) and repeat; if balanced, estimate and do inference. After matching, regression or weighting on the matched sample can correct residual imbalance.

Design choices.

- Which covariates to include (recall the adjustment criterion; never include post-treatment variables).

- Distance metric: Mahalanobis, propensity score, or hybrid.
- Number of matches M: fixed or varying (caliper-based). Smaller M: smaller bias, larger variance; larger M: the reverse. With a much larger control group, 1-to-many matching is natural.
- With or without replacement.

Mahalanobis distance accounts for correlations and scales: $d(x, y) = \sqrt{(x-y)^\top S^{-1}(x-y)}$ with S the sample covariance (treated group or pooled). With few continuous covariates ($p \lesssim 5$–8) it works well; with many covariates or a complex treatment model the PS gives effective dimension reduction. Common practice: match on the PS with a caliper, then refine with Mahalanobis distance within the caliper.

Caliper matching. With caliper width c, all controls within distance $< c$ of a treated unit are potential matches, so M varies by unit (e.g. $M_1 = 2$, $M_2 = 3$, $M_3 = 1$). A common choice is $c = 0.2 \times \mathrm{SD}(\mathrm{logit}\,\hat{e}(Z))$ (Austin 2011, simulation-based).

Full matching (Rosenbaum 1991; Hansen 2004). Standard matching discards unmatched controls. Full matching places every unit into a matched set with variable ratio (some 1:k, others k:1), minimising total within-set distance (solved efficiently by software). It uses more controls within the overlap and produces a weighted sample, a bridge between matching and IPW. Often combined with a caliper. In R: `MatchIt` with `method = "full"`.

With versus without replacement. With replacement: computationally easier, not order-dependent, but some units (especially with extreme PS) may be matched many times and dominate (like extreme IPW weights). Without replacement: avoids extreme reuse but is order-dependent (greedy vs. optimal matters). Software: `MatchIt` (Ho et al.), `Matching` (Sekhon).

Checking balance. Standardised mean difference of covariate k:

$$\mathrm{SMD}_k = \frac{1}{s_t} \cdot \frac{1}{N_1} \sum_{i=1}^{N} X_i \Big(Z_{ik} - \frac{1}{|\mathcal{M}_i|} \sum_{j \in \mathcal{M}_i} Z_{jk} \Big),$$

with s_t the SD of covariate k in the treated group before matching. Compare whole distributions via empirical CDFs and Kolmogorov–Smirnov statistics. **Do not use p-values** for balance: they conflate changes in balance with changes in sample size, and balance is an in-sample property (Ho et al. 2007).

Love plot. A dot plot of SMDs per covariate before and after adjustment, with reference lines at ± 0.1. In the LaLonde (1986) job-training data (propensity score, age, education, married, no degree, race indicators, earnings in 1974 and 1975), unadjusted SMDs range up to about 1.5–2 in absolute value; after matching they fall inside ± 0.1.

5 Model dependence and empirical evidence

Purpose. Explain why matching reduces sensitivity to modelling assumptions, and what the LaLonde benchmark teaches.

The extrapolation problem. When treated and control groups differ on covariates, regression must extrapolate into sparse regions. A linear and a quadratic specification can fit the control region equally well yet give very different values in the treated region. Model dependence is greatest where data are sparse (Ho, Imai, King and Stuart 2007).

Matching as preprocessing. After matching, the distributions overlap and regression interpolates rather than extrapolates; different specifications give similar estimates. Workflow: (1)

match for balance, (2) regress/weight on the matched sample, (3) obtain estimates robust to model choice. "Matching is the starting point, not the end."

The LaLonde benchmark. The National Supported Work (NSW) programme was a randomised job-training experiment (LaLonde 1986), with experimental ATT $\approx$ \$1,794. Replacing the experimental controls with observational ones (CPS/PSID), unadjusted estimates range from $-$\$15,000 to $+$\$2,000, because the observational controls differ dramatically in demographics, prior earnings and employment. It is the canonical testbed for matching methods.

Lessons. Dehejia and Wahba (1999): PS matching on a subsample with pre-treatment earnings recovers the experimental benchmark. Smith and Todd (2005): the result is fragile; different subsamples, matching methods and distance metrics lead to different conclusions. A deeper issue: the DW subsample included pre-treatment earnings, arguably the most important confounder; without it, ignorability itself is doubtful whatever the method. Key lessons: covariate sufficiency comes before method choice; matching can recover experimental benchmarks, but with no guarantees.

Practical takeaways.

- Matching is a nonparametric preprocessing step, not an end in itself; then apply regression or weighting on the matched sample.
- Always check multivariate covariate balance, not just PS balance.
- In high-dimensional settings combine matching with regression.
- Sensitivity analysis for unmeasured confounding is an important complement.

A ATE matching: why it is harder

For the ATE one must impute $\hat{Y}_i(0)$ for treated and $\hat{Y}_i(1)$ for control units. Each unit serves both as a subject and as a match for others, creating cross-group dependence absent for the ATT or ATC alone. Moreover the bias $O(N^{-1/p})$ dominates the $O(N^{-1/2})$ sampling error when $p > 2$: $\text{Bias}(\hat{\tau}_{\text{ATE}}) = O(N^{-1/p}) \gg O(N^{-1/2})$, so simple matching is not $\sqrt{N}$-consistent for the ATE. Use bias correction (Abadie and Imbens 2011) or IPW/DR instead (Abadie and Imbens 2006).

B Why the estimated PS can beat the true PS

The true $e(Z)$ is a population quantity; two units can have $e(Z_i) = e(Z_j)$ yet very different covariates. The estimated $\hat{e}(Z) = g(Z^\top \hat{\beta})$ is fit to this sample, so $\hat{e}(Z_i) \approx \hat{e}(Z_j)$ means Z_i, Z_j are close in the directions that distinguish treated from control in this sample: estimation noise acts as implicit covariate refinement. Under correct specification (Abadie and Imbens 2016), $\text{Var}(\hat{\tau}_{\text{ATT}} \mid \hat{e}) \leq \text{Var}(\hat{\tau}_{\text{ATT}} \mid e)$. Analogously, IPW with $\hat{e}$ attains the semiparametric efficiency bound while IPW with e does not (Hirano, Imbens and Ridder 2003). The advantage disappears under misspecification.

C The PS paradox (King and Nielsen 2019)

Many covariate configurations map to the same PS. Unit A (treated) with high income and low education and unit B (control) with low income and high education may both have $\hat{e} = 0.50$: perfect PS balance, opposite covariates. A tighter PS caliper yields more "PS-similar but

covariate-different" pairs, so a better PS match can mean worse covariate balance; the remaining imbalance may require more extrapolation than the raw data, increasing model dependence.

D PS overfitting and the broader picture

Overfitting the PS model can help: the goal is in-sample balance, and a flexible model captures sample-specific patterns so that similar $\hat{e}$ means closer Z. The limit is overlap: if $\hat{e} \to 0$ or 1 for all units, overlap is destroyed. Three results, one message: the estimated PS beats the true PS (sample adaptation), overfitting helps (the same mechanism), but the PS paradox shows that 1D compression can still lose information. The PS is a dimension-reduction tool, not a substitute for checking balance. Practical resolution: use the PS for dimension reduction (especially for large p), always verify multivariate balance, and consider Mahalanobis matching within a PS caliper.

E Beyond binary treatment

With multiple treatment levels (doses, programmes): compare pairs by standard binary matching, or use the generalised propensity score (GPS) for K levels (Imai and van Dyk 2004). For continuous treatments the GPS is the conditional density of treatment given covariates (Hirano and Imbens 2004). In practice subclassification or weighting on the GPS is more common than matching.

F Bias-corrected matching versus post-matching regression

In bias-corrected matching, regression is a correction term inside each matched pair; matching is primary and regression fixes the residual $Z_i \neq Z_j$ gap. In post-matching regression, matching is preprocessing (balanced sample) and a regression $Y \sim X + Z$ on the matched data does the estimation. Why not regress on the full sample? $\hat{\mu}_0$ can be learned from all controls, and balance is irrelevant if the model is correct; but if the model is wrong, full-sample regression extrapolates and is model-dependent.

	Model correct	Model wrong
Full-sample regression	optimal (more data)	risky (extrapolation)
Post-matching regression	less efficient (data discarded)	safe (interpolation)

If controls lie in $Z \in [0, 3]$ and treated in $[5, 8]$, full regression must extrapolate $\hat{\mu}_0(6)$: a linear model gives answer A, a quadratic answer B. After matching, controls also lie near $[5, 8]$ and both models give similar answers. This is the core of "matching as preprocessing" (Ho et al. 2007).

G Stratification

With a single discrete covariate Z with levels k that makes assignment ignorable, $\mathbb{E}[Y(1)] = \sum_k \mathbb{E}(Y \mid Z = k, X = 1)P(Z = k)$. With n_k units in cell k and cell means $\bar{Y}_{k,x}$, estimate $\mathbb{E}[Y(1)]$ by $\sum_k \bar{Y}_{k,1} n_k/n$, hence

$$\hat{\tau} = \sum_k (\bar{Y}_{k,1} - \bar{Y}_{k,0}) \frac{n_k}{n}.$$

For continuous Z, split it into k classes: $\hat{\tau}_k = \sum_k (\bar{Y}_{k,1} - \bar{Y}_{k,0}) n_k / n$, generally biased for τ. Let $R_k = 1 - \frac{\mathbb{E}(\hat{\tau}_k) - \tau}{\mathbb{E}(\hat{\tau}_1) - \tau}$ be the fraction of bias removed by k-stratum stratification. Cochran (1968): $R_k \geq 90\%$ for $k \geq 5$ across a broad class of models, so stratifying into five or more blocks removes roughly 90% of the confounding bias.

H Source coverage

The source deck has 56 physical pages (47 numbered slides).

Topic	Physical pages
Setup, regression vs. matching, design, estimator, exact/CEM	1–11
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